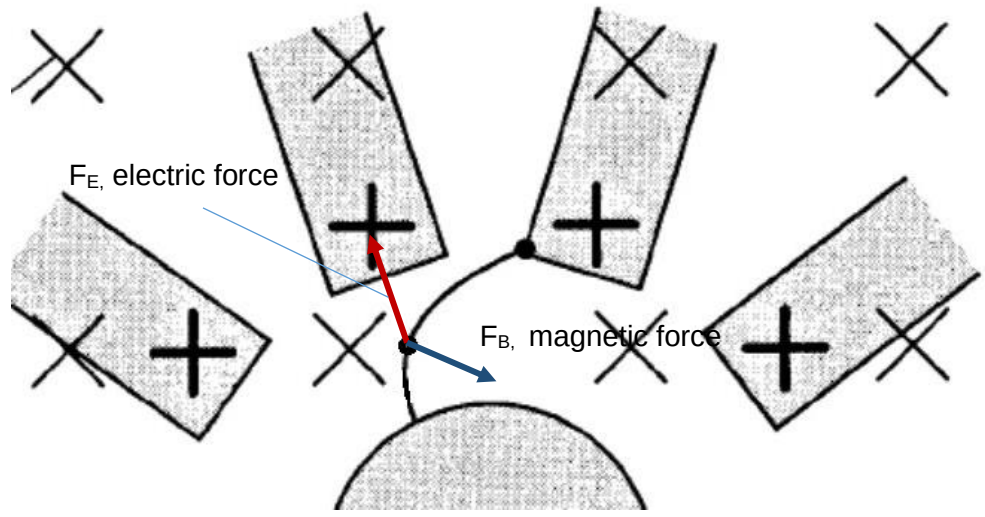


7ai <i>para</i> <i>2</i>	To <u>prevent microwave/(em) radiation</u> from <u>leaking/escaping out/exiting</u> of the cage/microwave oven.	E	A1
7aii	From the passage, by international convention, microwave ovens operate at frequencies at around 2.45 GHz. Wavelength of the microwaves = <u>$3.00 \times 10^8 / 2.45 \times 10^9 \text{ Hz} = 0.122 \text{ m}$</u>	A	C1

para 1, 2 & 5	100 times smaller than 0.122 m or 12.2 cm = 1.22×10^{-3} m or 0.00122 m. Hence, estimated spacing of holes is <u>1×10^{-3} m</u> or <u>1.2×10^{-3} m</u> . [-1] Powers of Ten error for conversion of GHz		A1
7bi para 3	Potential difference across the electrodes = 5000.00 V By conservation of energy, <u>Kinetic energy gained by an electron = electrical potential energy loss by the electron</u> $= (5000.00)(1.60 \times 10^{-19}) = 8.00 \times 10^{-16}$ J $= 8.0 \times 10^{-16}$ J	E	M1 A0
7bii para 3	Power output, $P = (\text{energy of an electron}) \times (n/t)$ Each electron has available max. 8.0×10^{-16} J of energy (assuming that the electrons start off from cathode with negligible kinetic energy) to be converted to microwaves. Least number of electrons per second = $\frac{1000}{8.0 \times 10^{-16}} = 1.25 \times 10^{18} \text{ s}^{-1}$	A	A1

7biii para 3	Not all the (kinetic) energy of the electrons is converted into microwave energy as: - electrons give off e.m. radiation of varying wavelengths as it accelerates towards the anode and hence actual energy possessed by electrons are lower when reaching the anode - some electrons hit the anode, some of its kinetic energy is also converted to thermal energy / passed to the molecules (or atoms) in the anode causing thermal agitation so less energy is available for conversion to microwave energy <u>Not all the microwaves generated from the energy is fed into the cavity resulting in energy losses</u> due to: - the walls in the cavity of the food chamber absorbing some of the microwaves - microwaves may be fed back / coupled back to the magnetron (resulting in actual useful power of microwaves less than the actual energy that can be supplied by the electrons) Answer must be related back to the efficiency of <u>conversion of energy of the electrons to the power output of the microwaves; or loss in microwaves produced fed into cavity.</u>	D	A1
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7biv 1 para 3	 Correct direction of F_E drawn and labelled. (// to electric field line towards anode, see Fig. 7.3 for the electric field between the anode and cathode.) Correct direction of F_B drawn and labelled. (The velocity of the electron should be tangential to the trajectory. Apply Fleming's left hand rule to get direction, towards centre of "circular motion" and perpendicular to trajectory) <i>[-1] if F_E and F_B are not defined.</i> <i>Ignore weight if labelled. (Weight is insignificant to electric and magnetic forces)</i>	A A1 A1
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7biv 2 para 3	As the magnetic force on the electron is acting perpendicular to its motion, (no work is done by the magnetic force on the electron), hence it does not affect the speed of motion as it moves towards the anode. Hence the maximum kinetic energy gained remains unchanged.	D B1 A1																		
7ci para 6	<table border="1" data-bbox="279 1332 1308 1691"> <thead> <tr> <th>depth into food z / mm</th><th>intensity of microwaves at depth z $I / \text{A.U.}$</th><th>$\ln (I / \text{A.U.})$</th></tr> </thead> <tbody> <tr> <td>0</td><td>24</td><td>3.18</td></tr> <tr> <td>4</td><td>19</td><td>2.94</td></tr> <tr> <td>8</td><td>15</td><td>2.71</td></tr> <tr> <td>12</td><td>12</td><td>2.49</td></tr> <tr> <td>16</td><td>10</td><td>2.30</td></tr> </tbody> </table>	depth into food z / mm	intensity of microwaves at depth z $I / \text{A.U.}$	$\ln (I / \text{A.U.})$	0	24	3.18	4	19	2.94	8	15	2.71	12	12	2.49	16	10	2.30	E A1 A1
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7cii 1. 2. para 6	1. Plot the point accurately to $\frac{1}{2}$ smallest square. 2. Appropriate best fit line drawn.	E A1 A1
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7cii para 6	Using (0, 3.16) and (15.2, 2.32), $\text{Gradient} = \frac{3.16 - 2.32}{0 - 15.2} = -\frac{0.84}{15.2} = -0.055$ <i>Correct computation of the gradient with two points on the best fit line.</i> <i>[-1] if the points are not read to $\frac{1}{2}$ smallest square precision, $\frac{1}{2}$ square off. If more than one square off [0] for whole part.</i> <i>[-1] if the two points chosen are less than half the line drawn</i> <i>No units are required. However, if units are incorrect [-1]</i>	E C1 A1
7civ para 6 & 7	$I = I_0 e^{-\mu z} \Rightarrow \ln I = \ln I_0 - \mu z$ Hence, gradient of the graph = $-\mu$ When $z = \delta_p$, $I = I_0/e$ $\Rightarrow \frac{I_0}{e} = I_0 e^{-\mu \delta_p} \Rightarrow \delta_p = 1/\mu = 1/0.055 = 18 \text{ mm}$	D C1 A1
7cv1 para 5	The potato mash with a higher water content would have <u>more of the microwave (energy) absorbed at the surface/ more microwaves absorbed per unit length/per unit volume by the water molecules</u> as the microwaves move through the food, (the intensity of the microwaves will then fall to 1/e of its intensity at the surface in a shorter depth), causing the penetration depth to be <u>smaller</u>.	D M1 A1
7cv2	N – straight line graph with y-intercept unchanged, gradient is steeper. <i>Graph must be coherent with conclusion of penetration depth obtained earlier.</i>	E A1

<i>para</i> 6			
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